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Fiber-reinforced polymer (FRP) rockbolts and their potential application in low pH mines: A state-of-the-art literature review

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ABSTRACT

This review paper explores the mechanisms and factors influencing the corrosion of rockbolts, particularly in aggressive underground mining environments characterized by high humidity, fluctuating temperatures, and corrosive chemical agents. The study also highlights the role of chloride ions and sulfide inclusions in initiating and propagating corrosion. Additionally, the review explores the emergence of fiber-reinforced polymer (FRP) rockbolts as a viable alternative, offering insights into their non-corrosive properties, lightweight nature, and high tensile strength. The comparative analysis between traditional steel rockbolts and FRP alternatives reinforces the potential for FRP rockbolts to enhance the longevity and safety of mining support structures, while also considering the practical implications of their implementation in the mining industry.

1. Introduction

Rockbolts are essential for stabilizing rock masses in mining and tunneling projects, playing a crucial role in preventing collapses and ensuring overall stability [1]. They work by transferring the load from unstable zones to the more stable, confined interior of the rock mass [2]. The versatility of rockbolts allows them to be effective in various geological conditions, making them an adaptable solution for different rock formations [1,3].

Rockbolts are typically made of steel, a material that is prone to corrosion in acidic environments [4]. Low pH conditions facilitate the electrochemical reactions between the steel and the surrounding environment, resulting in the formation of iron ions and hydroxide ions [5]. The acidic environment can be a result of natural geological processes or human-induced activities. For instance, the presence of acidic groundwater in massive sulfide deposits or the use of aggressive chemicals in mining operations can contribute to lowering the pH levels. Moreover, the byproducts of mining, such as sulfide minerals, can react with water and oxygen to produce sulfuric acid, further exacerbating the corrosion problem [6–8]. Corrosion can lead to reduced strength and a shorter lifespan for the steel rockbolts, compromising their ability to provide effective ground support [9,10]. These limitations necessitate the use of more corrosion-resistant materials or protective coatings to ensure the safety and stability of mining operations [11].

Fiber-reinforced polymer (FRP) rockbolts offer a robust solution for reinforcing mine openings in corrosive environments [12]. FRP rockbolts, which are made from materials like fiberglass bonded with polymers such as vinyl ester, or epoxy, are corrosion-resistant [13]. They have gained attention for their use as permanent rock support in civil engineering infrastructure like roads and tunnels due to their lightweight, lower environmental impact, and excellent durability characteristics [14,15]. While FRP rockbolts have been primarily used as temporary support in mining operations, their potential use as permanent support is increasingly recognized [16]. In low pH mines, the use of FRP rockbolts can significantly improve safety and reduce maintenance costs associated with the monitoring and replacement of corroded rockbolts. Their adoption would not only ensure the structural stability of the mine but would also promote a safer working environment for mine operators. As mining operations continue to encounter challenging geological conditions, the importance of corrosion-resistant rockbolts like FRP becomes ever more critical [17,18].

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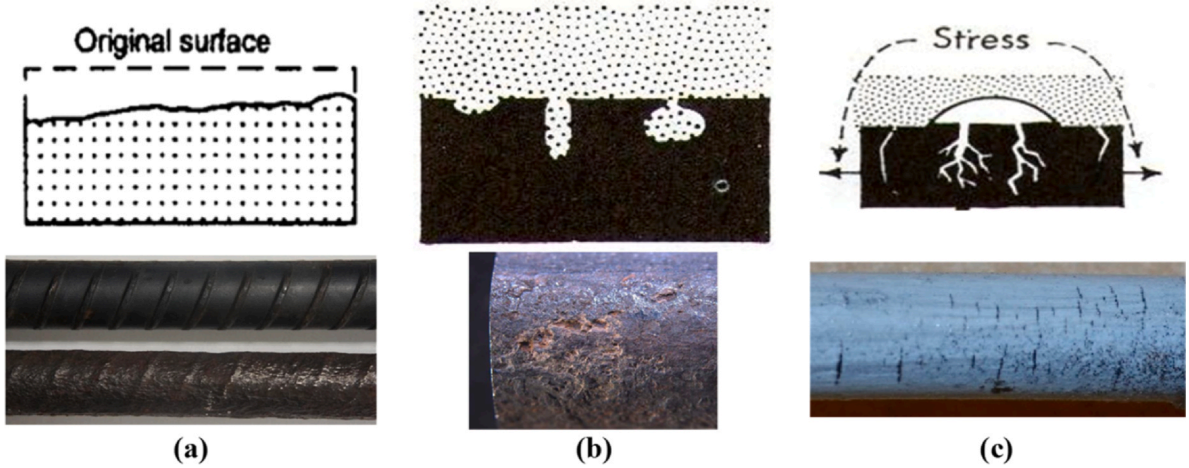

Fig. 1. Corrosion mechanism in steel rockbolts: (a) uniform corrosion; (b) pitting corrosion; and (c) stress concentration cracking. Notes: sub-figures from top-to-bottom are schematic representation of corrosion (after Aziz et al. [31]) and corrosion examples (after Aziz et al. [37] and Wu et al. [35]).

This review paper presents the factors influencing rockbolt corrosion, particularly in low pH environments. Moreover, it highlights the emergence of FRP rockbolts as a promising alternative to steel rockbolts. The comparative analysis between traditional steel rockbolts and FRP alternatives reinforces the potential of FRP rockbolts to increase the service life and safety of mine workings. Additionally, the study identifies the research gaps and provides recommendations for future research on FRP rockbolts in the mining industry.

2. Corrosion problem in mines

Corrosion in mines is influenced by several key factors, each contributing to the degradation of mining equipment and infrastructure [19]. High humidity, fluctuating temperatures, and the presence of corrosive agents like chloride and sulfate ions in mine water accelerate corrosion [20]. The pH environments in mines can vary substantially, often influenced by the presence of acid mine drainage (AMD) [21], also referred to as acid rock drainage (ARD) [21,22]. AMD is the highly acidic leachate that is often contaminated with dissolved metals and sulfates [7]. The primary cause of low pH in mine environments is the oxidation of sulfide minerals, most notably pyrite [23]. When sulfide minerals are exposed to oxygen and water, a series of reactions occur, leading to the production of sulfuric acid, which drastically lowers the pH of the mine environment [7,21,24].

One of the main sources of AMD is mine waste rock dumps. These are piles of rock removed during mining that contain sulfide minerals. When rainwater and atmospheric oxygen come into contact with these rocks, oxidation occurs, generating acidic runoff. Similarly, tailings impoundments, which contain the finely ground waste material left over after ore processing and are often rich in sulfides, are significant sources of AMD. In underground mines, sulfide minerals in exposed rock faces and fractured zones can oxidize, and the resulting acidic water can accumulate and eventually discharge. Open pit mines expose vast amounts of rock to oxidizing conditions, and the pit walls and stockpiled ore can generate AMD that collects in the pit or flows out as seepage [21].

Moreover, pumped or naturally discharged underground water can be acidic and contain high concentrations of metals, constituting a direct source of AMD into the environment. Diffuse seeps from replaced overburden in rehabilitated areas can also contribute to AMD if the backfill material contains sulfide minerals that were not adequately treated or isolated. Even construction rock used in the construction of roads, dams, etc., can become a source of AMD if it contains reactive sulfide minerals [21]. Abandoned and closed mine sites continue to generate AMD for decades due to the ongoing oxidation of exposed sulfide minerals. The collapse of old mines can create sinkholes that allow surface water to interact with sulfide-bearing material underground, leading to AMD contamination of both groundwater and surface water [25]. Activities such as blasting, drilling wells, and excavating for infrastructure can disturb sulfide-bearing bedrock, exposing new surfaces to air and water and initiating or exacerbating ARD [22].

Microorganisms, particularly acidophilic bacteria such as *Acidithiobacillus ferrooxidans* (*A. ferrooxidans*), play a crucial role in catalyzing and accelerating the oxidation of ferrous iron to ferric iron and the sulfide minerals themselves [7,24,26]. These biological processes can increase the rate of acid generation by several orders of magnitude compared to abiotic (non-biological) reactions [24]. *A. ferrooxidans* thrives in acidic environments, with optimal activity typically occurring at pH values below approximately three [27]. This microbial activity contributes significantly to the extremely low pH values observed in many AMD sites [24].

The low pH of AMD makes it inherently corrosive to many materials commonly used in mine structures, including metals and concrete [22]. Sulfuric acid, a primary component of AMD, reacts aggressively with these materials, leading to their degradation [7]. Steel support systems in underground mines, such as cable bolts and rock bolts, are susceptible to corrosion in mine environments [26,28]. Mine mesh, used for ground support, is also vulnerable to corrosion [19]. The study indicates that wet-dry cycles and the presence of dust in underground mines can increase the corrosion rates of steel mesh.

2.1. Corrosion mechanism in steel rockbolts

Field and laboratory tests were carried out to investigate how rockbolts corrode and respond to different conditions [29–36]. The typical mechanisms of corrosion that significantly affect steel rockbolts are shown in Fig. 1 and include: (a) uniform corrosion, which occurs evenly across the surface of the rockbolt, leading to a gradual thinning of the material. Uniform corrosion is associated with acidic aqueous corrosion and atmospheric corrosion. (b) Pitting corrosion is a localized form that creates small pits or holes in the rockbolt, significantly weakening it and potentially leading to sudden failure. (c) Stress corrosion cracking is a combination of tensile stress and a corrosive environment, leading to the formation of cracks.

2.2. Low pH level affecting steel rockbolts

The corrosion of steel rockbolts in acidic conditions is well-documented in the literature as an aggressive corrosion regime [5,38]. Fig. 2

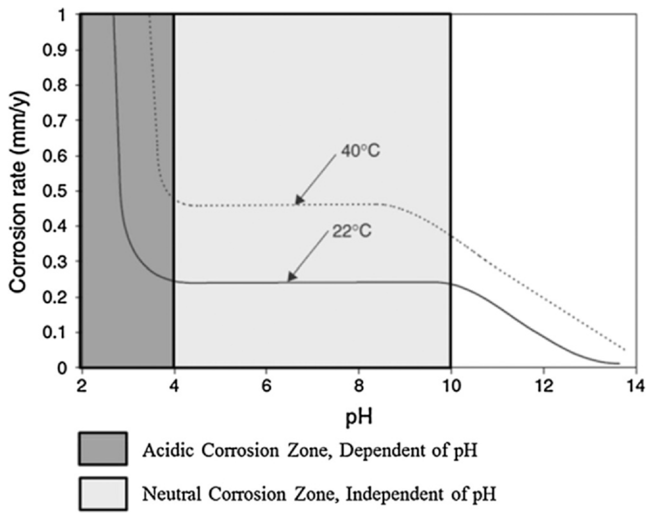

Fig. 2. Corrosion rate of steel as a function of pH in water containing 5 ppm of dissolved oxygen at 22 °C and 40 °C [38].

shows that the corrosion rate increases dramatically as the pH decreases. Below a pH of 4, steel corrosion rates rise from 0.25 mm/year to over 1 mm/year at 22 °C. The process is independent of dissolved oxygen, but is significantly influenced by temperature, with corrosion rates approximately doubling when the temperature increases from 22 to 40 °C.

In certain geo-environments, the corrosion problem for rockbolts becomes so severe that it can lead to reinforcement failure [32]. Highly corrosive ground conditions have been reported in some hard rock mines in the United States, causing expandable rockbolts to corrode within just a few months after installation. The pH of hard rock mine water can range from acidic (pH of − 3.5 to 5) to circumneutral (pH of 5–8), depending on the presence of sulfides, the geological features of the mine, and the presence of carbonate minerals [6,39]. Unexpected rockbolt failures can lead to roof falls and interrupted mining production, forcing mine operators to undertake expensive and strenuous rehabilitation on a regular basis due to safety hazard concerns [32].

In another study, the corrosion of rockbolts in the Sanshandao Gold mine, a coastal underground mine in Jiaodong Peninsula, China was examined [40]. Groundwater in the mine is rich in chloride ions and has a pH range of 4.68–7.14, indicating conditions that range from acidic to circumneutral. These factors contribute to the corrosion of metal rockbolts, which are crucial for maintaining the stability of underground excavations. The corrosion rate of rockbolts in the mine was estimated to be from 0.5 to 1.5 mm/year. As corrosion progresses, the anchoring force of the rockbolts decreases, potentially compromising the stability of the mine opening [40].

Several strategies have been presented to protect steel rockbolts in low pH environments, such as coatings, galvanized steel, and stainless-steel alternatives [41]. There is a perception that there is no corrosion issue with fully encapsulated resin-grouted bolts. However, this is incorrect because resin shrinks upon curing [42]. Moreover, resin can be damaged under shear and tensile loading, exposing unprotected steel and allowing corrosion. Zinc galvanisation on steel rockbolts provides excellent corrosion resistance in alkaline conditions; however, it performs poorly in acidic environments [11]. The zinc coating can become ineffective or vulnerable due to micro-cracking or splitting during bolt inflation, or scratching during rockbolt insertion. Stainless steel can be highly resistant to corrosion, increasing the service life of the bolts. However, it tends to be costly and is not commonly used in mines [41]. Therefore, there is a need for non-corrosive and cost-effective alternatives, such as fiber-reinforced polymer (FRP) rockbolts.

3. FRP rockbolt

FRP rockbolts are made of a composite material consisting of continuous fibers (glass, carbon, basalt, or aramid) embedded in a polymer resin matrix, typically a thermoset resin like polyester, vinyl ester, or epoxy. Fibers provide high tensile strength, while the resin matrix binds the fibers together and protects them from environmental damage [13]. Choosing the right FRP material depends on the specific application requirements, including the necessary strength and stiffness, the corrosion environment it will be exposed to, cost considerations, and the material's durability and fatigue resistance. Each of these factors plays a crucial role in determining the most suitable FRP material to ensure optimal performance and long service life.

Resins are crucial in defining the properties of FRP rockbolts by binding fibers, transferring stress, and providing environmental protection. They significantly influence strength, stiffness, toughness, and fatigue resistance. Epoxy resins offer the best mechanical properties, followed by vinyl ester and polyester. The chemical resistance varies, with vinyl ester and epoxy outperforming polyester. Temperature resistance is highest in epoxy, then vinyl ester, and polyester. The long-term durability against ultraviolet light and moisture is influenced by the resins. Epoxy and vinyl ester resins performed better than polyester resin, providing long-term durability [43]. Therefore, selecting the right resin system based on mechanical, chemical, and thermal requirements, as well as cost, is vital.

Glass fiber is the most common and economical option, offering high tensile strength and good chemical resistance. Glass fibers used in rockbolts include E-glass (providing good electrical and mechanical properties), S-glass (offering higher tensile strength than E-glass), EC-R glass (an E-glass variant with enhanced corrosion resistance), and AR-glass (providing improved alkali resistance) [14]. Newer fibers, such as basalt fiber, exhibit good strength and chemical resistance and are potentially more sustainable. Carbon fiber offers high strength and stiffness compared to other fibers but can be more expensive and electrically conductive, which may be a concern in some applications. Aramid fiber has high toughness and impact resistance, however, lower compressive strength compared to other fibers and is susceptible to ultraviolet degradation [43]. A study on the environmental impact on the durability of FRP bars showed that the fibers should be mixed with the corresponding resin matrix to achieve good durability. Glass fiber with vinyl ester and basalt fiber with epoxy were found to have excellent durability. Additionally, the GFRP bars provide better corrosion resistance compared to other FRP bar types in acid solutions [44].

Pultrusion, as shown in Fig. 3, is a prevalent manufacturing technique for FRP rockbolts [45]. This process is optimal for the continuous fabrication of composite parts with a consistent cross-sectional profile. Initially, fibers, such as glass fibers are packaged in roving, then drawn through a resin bath where the material is thoroughly impregnated with a liquid thermoset resin, such as epoxy. The fibers are spread out to ensure uniform wetting prior to entering the resin bath. The resin-impregnated fibers are then directed through a heating die that defines the final dimensions of the bar. Subsequently, helicoidal wraps are applied to the bar's surface before it enters a curing oven. Within the oven, heat is precisely controlled at approximately 170 °C, facilitating the curing of resin and its transformation from a liquid to a solid state. The solid bar exits the curing oven, conforming precisely to the die cavity's dimensions. The bar is continuously pulled and finally, cut to the desired length. The duration of this process varies depending on the size of the final bar, with a typical production speed of approximately 1 m per minute [14,43]. Pultrusion is advantageous for producing high-strength, lightweight, and consistent profiles, making it efficient for mass production with precise control over the fiber-resin ratio. The commonly available glass FRP (GFRP) and basalt FRP (BFRP) rockbolts are shown in Fig. 4.

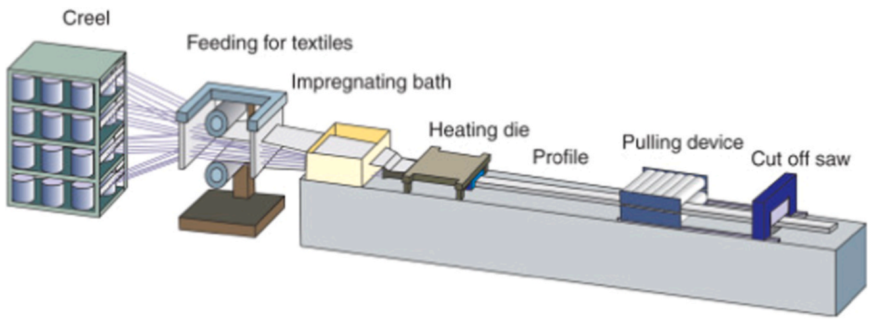

Fig. 3. FRP rod manufacturing process- pultrusion [45].

Fig. 4. Fiber-reinforced polymer (FRP)- rockbolt [46].

3.1. Properties

Table 1 lists the properties of a GFRP rockbolt alongside those of a standard 20 mm diameter steel rockbolt. FRP rockbolts exhibit several key mechanical properties, including high tensile strength, often comparable to or exceeding that of steel rockbolts, providing excellent support and reinforcement to rock structures. They have a lower modulus of elasticity compared to steel, resulting in greater flexibility [47]. Additionally, FRP rockbolts are much lighter than steel rockbolts, making them easier to handle and install. They also demonstrate excellent fatigue resistance, making them suitable for long-term applications in dynamic environments [48]. Moreover, FRP rockbolts are highly resistant to corrosion, enhancing their durability in harsh conditions such as low pH mines.

The tensile behavior of FRP composed of a single type of fiber material is characterized by a linear elastic stress-strain relationship until failure. Testing the tensile strength of FRP rockbolts is challenging due to stress concentrations around the anchorage points, which can cause premature failure. Therefore, it is crucial to use a testing grip that ensures failure occurs in the middle of the test specimen [49]. Gilbert et al. [50] examined the impact of short FRP bolt samples on tensile strength. FRP sample sizes were 500 and 1200 mm in length with ends embedded in 150- and 450-mm long steel tubes, respectively, for anchoring to the testing machine jaws. The tensile tests showed consistent tensile strength regardless of the FRP bolt sample length. However, the 1200 mm sample exhibited a 50 % increase in elongation, necessitating further tests to confirm this finding. It is recommended to use the ASTM D7205/D7205M [51] test methods to determine the tensile strength and stiffness of FRP rockbolts accurately.

Table 1
Mechanical properties of GFRP [46] and steel rockbolt [16].

Properties	GFRP rockbolt	Steel rockbolt
Diameter (mm)	21	20
Cross-sectional area (mm^2)	350	299
Tensile strength (MPa)	1000	600
Elastic Modulus (GPa)	60	200
Density (kg/m^3)	2200	7800
Strength-to-weight ratio (MPa/MN/m^3)	46,350	7844

The double-shear capacity of FRP rockbolts at rock joints is a concern due to the brittle nature of FRP materials under shear failure. It is essential to determine the shear strength of FRP bolts based on simulated ground conditions, making it logical to test bolts in rock or cementitious mediums [52]. The loading system was designed based on published studies [53], as a standard double-shear test procedure for bolted joints is rarely specified. The concrete block used was 200 × 200 × 200 mm, as shown in Fig. 5. The BFRP cable was embedded in the joint either perpendicularly [52] or at inclinations of 0°, 5°, 10°, and 15° [54]. Resin grout served as the bonding material, with external fixtures including an M60 nut, a face plate, and a bonded anchorage. The concrete was cured for 28 days before testing. A vertical load was applied at a controlled rate of 1 mm/min using an electro-hydraulic servo universal testing machine to induce shear displacement along the rock joint. It was observed that BFRP rockbolts exhibited a shear failure mode, while steel bolts showed ductile deformation under combined shear and bending forces. For non-pre-tensioned specimens, BFRP rockbolts reached ultimate strength and failed with less deformation compared to steel bolts. However, pretension load improved the shear capacity of BFRP rockbolts. A 70 kN pretension increased shear capacity by 81.6 % and prevented early debonding of the grout-bolt interface [52]. Additionally, the ultimate shear capacity increased with the inclination angle of the BFRP rockbolts. At a 15° inclination angle, the ultimate shear capacity was 51 % higher than at 0° [54]. Overall, pretension load and inclination angle proved to be critical factors in improving the performance of BFRP rockbolts, though their brittleness requires careful design considerations for underground applications.

The pullout test is a common method for evaluating the performance of FRP rockbolts under axial loading. When anchoring a steel rod in concrete, the bond force can be transferred through adhesion resistance, frictional resistance at the interface, and mechanical interlock due to interface irregularities. In contrast, for FRP, it is believed that the bond force is transferred through the resin to the reinforcement fibers, with bond-shear failure in the resin being a possibility [55]. The pullout test setup, adapted from BS 7861-1 [56], is shown in Fig. 6 and consists of a concrete block, resin grout, cable, and bonded anchorage. The load step was 5 kN, and each load interval was sustained for data recording. BFRP rockbolts exhibited two types of failure: debonding at shorter encapsulated lengths and rupture at longer encapsulated lengths. Steel bolts, however, underwent yielding and sustained the load beyond this threshold. BFRP bolts had lower initial pullout stiffness than steel bolts due to their lower modulus of elasticity. However, BFRP bolts achieved higher pullout strength compared to steel bolts [52]. Another study reported no difference in pullout stiffness or grout-bar bond strength between steel and GFRP rockbolts [57].

FRP rockbolts are anisotropic in nature and their characteristics are influenced by factors such as fiber volume, types of fiber and resin used as discussed earlier. Mechanical properties are reduced if the fiber content is less than 70 % of the weight [14,58]. The weight gain of FRP conditioned in water and alkaline at 80 °C is approximately three times higher than at 20 °C [59]. Generally, FRP experiences mass loss

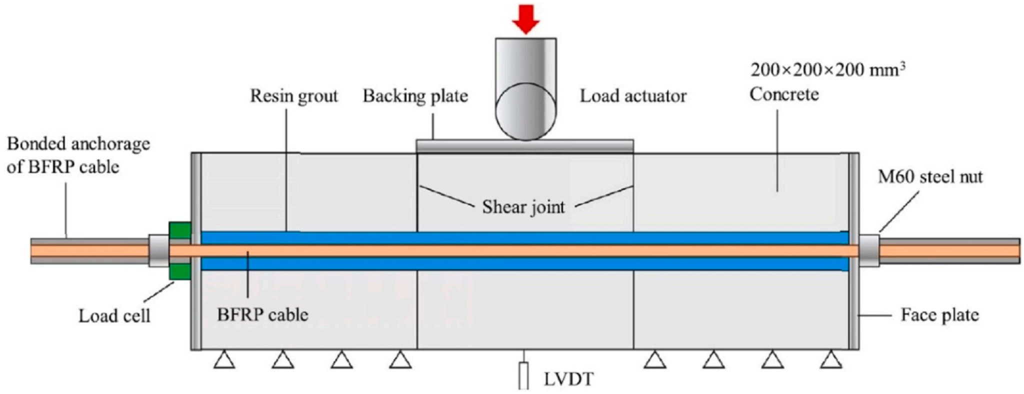

Fig. 5. Experimental setup for double-shear test on rock joint [52].

between 200 and 450 °C [60]. The variation in mass with temperature was measured by thermogravimetric analysis (TGA) [61]. A sharp decline in FRP mass, primarily between 300 and 450 °C, results in an 18 % drop due to the thermal degradation of the polymer [60,62], as shown in Fig. 7.

The mechanical properties of FRP are highly sensitive to temperature, primarily due to the degradation of the resin matrix. When the ambient temperature exceeds the resin's glass transition temperature (T_g), approximately 120 °C, the resin softens and cannot efficiently transfer stress between the fibers, reducing the synergistic effect between the fibers and the resin matrix [55]. ASTM D7957/D7957M [63] recommends a minimum T_g of 100 °C for FRP reinforcement. As shown in Fig. 8**(a)** and **(b)**, the tensile and shear properties of FRP bars remain unchanged from − 40 °C to 50 °C [62]. Below − 50 °C, the mobility of the polymer's molecular chains decreases, increasing the mechanical stress required to rupture the material. Above T_g, the breaking of molecular bonds begins, increasing the material's ductility and decreasing its mechanical strength and stiffness [60,62]. At temperatures above 300 °C, significant degradation of the polymer occurs (such as combustion and oxidation), severely reducing the load transfer provided by the matrix [60,64].

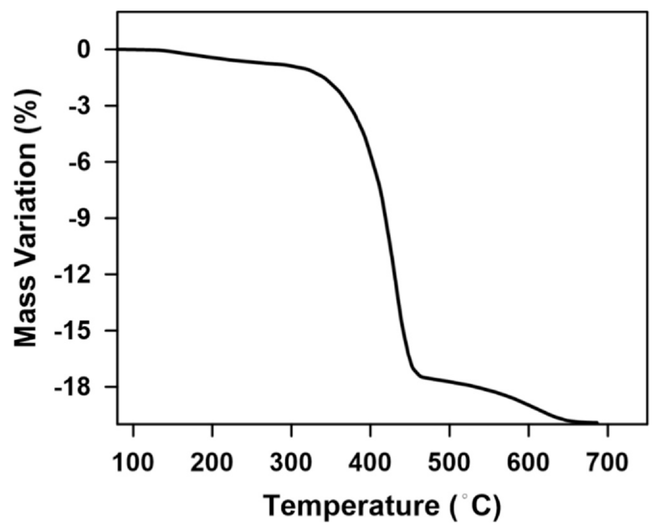

Fig. 7. Mass variation of GFRP bar specimen when heated between 20 and 800 °C [62].

3.2. Installation method

The FRP rockbolt consists of a threaded bar, spin bolt, nut, and a plate as shown in Fig. 9. The spin bolt, nut, and plate system are employed to enhance the installation process, particularly in hard coal mines. This FRP rockbolt system is engineered to expedite and pretension the installation of bolt groups during systematic bolting, thereby providing advanced strata support in mine galleries. The nut is designed to facilitate initial torque for spinning in, allow free running for tensioning, and protect against over-spinning of the bolt shaft. This results in a safe, efficient, and highly cost-effective installation procedure, significantly improving the overall stability and support within the mining environment [14].

FRP rockbolts, while serving the same ground support function as steel rockbolts, require specific installation considerations due to their distinct material properties. Drilling procedures are similar, but with FRPs, care must be taken to minimize borehole wall damage that could compromise grout bonding. Rockbolt insertion demands careful handling to prevent surface damage like scratches or impacts, which can significantly reduce FRP strength. Grouting remains crucial for load

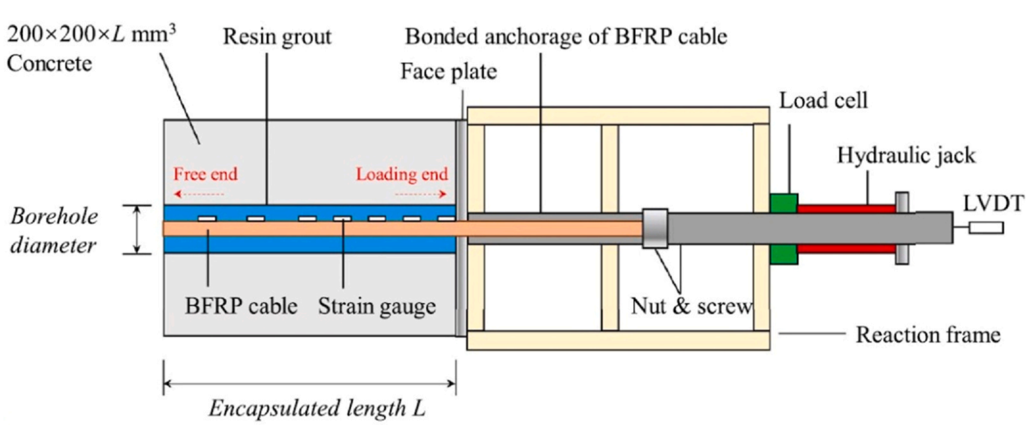

Fig. 6. Experimental setup for pullout test of FRP rockbolt [52].

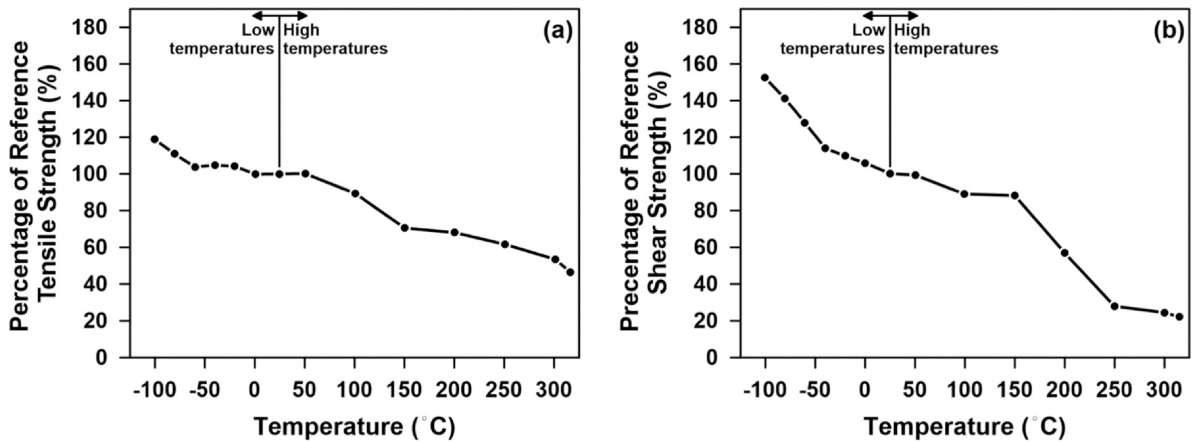

Fig. 8. Normalized average value of GFRP specimens tested under different temperatures: (a) tensile; and (b) shear strength [62].

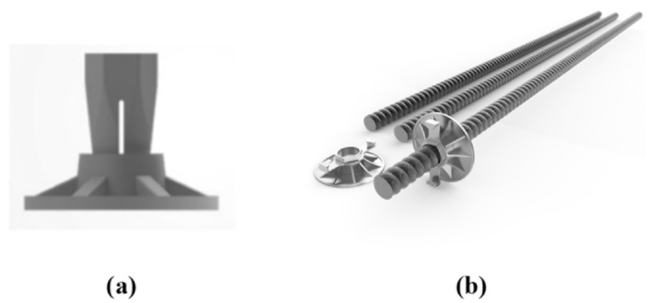

Fig. 9. (a) Spin bolt, nut and plate; and (b) FRP rockbolt with a plate [46].

transfer and environmental protection, with cementitious grouts, resin capsules, or combined systems employed. However, tensioning FRP rockbolts requires greater control to avoid over-tensioning and potential damage, unlike the more robust steel bolts [65]. Moreover, while FRP's corrosion resistance is a major advantage, proper grouting is still vital to prevent water ingress and ensure long-term performance. Overall, installing FRP rockbolts demands careful handling and controlled tensioning to ensure optimal bonding to maximize their performance in ground support applications.

3.3. Life-cycle assessments of FRP rockbolts

The lightweight nature of FRP bolts results in lower transportation emissions compared to conventional steel bolts, which emit significant amounts of carbon dioxide during manufacturing. This was confirmed by the life-cycle assessment (LCA) conducted on rockbolts by Kodymova et al. [66]. The LCA encompassed the entire life cycle of the product, including production, transportation, installation, and maintenance phases. The study used a one-meter rockbolt with a 100-year lifespan as the functional unit, comparing galvanized steel bolts and FRP rockbolts.

The transportation phase considered two options: shipping from China to Europe or local trucking within Europe. The installation phase involved drilling a borehole and installing the bolts with polyester resin capsules or cement mortar, assuming both bolts were anchored with resin cartridges. Environmental impacts were assessed across ten categories: abiotic depletion, acidification, eutrophication, global warming, ozone depletion, human toxicity, freshwater aquatic ecotoxicity, marine aquatic ecotoxicity, terrestrial ecotoxicity, and photochemical oxidation.

The study results indicated that FRP bolts have an environmental impact of approximately 15–65 % of that of the steel bolts, depending on the impact category, as shown in Fig. 10. For instance, the environmental impact related to global warming is about 40 % for FRP bolts compared to steel bolts. These findings confirm that FRP rockbolts are more environmentally friendly.

4. Application of FRP rockbolts

FRP rockbolts have found valuable applications in both mining and tunneling, providing effective ground support solutions in diverse and often challenging environments [49,67]. In mining, their corrosion resistance is particularly advantageous in operations dealing with acid mine drainage or high salinity, ensuring long-term stability of excavations and reducing maintenance costs associated with corroded steel bolts. Moreover, they offer reliable support without the risk of degradation, crucial for worker safety and preventing costly collapses. In underground coal mines with high gas levels, steel bolts in the mined rib need to be removed beforehand to avoid frictional sparks during the shearer cutting process. FRP rockbolts are a perfect alternative for rib support in coal mines because they can be cut by a shearer without producing sparks [49]. Studies have shown that using FRP rockbolts for rib support maintains the integrity of the surrounding rock effectively [68]. For mine development applications such as drifts and ramps, FRP bolts should provide stable and safe passageways [18]. Similarly, in civil engineering tunneling projects for highways, railways, and subways, FRP rockbolts offer crucial ground support, with their corrosion resistance being particularly beneficial in humid environments or areas exposed to de-icing salts [16].

4.1. Temporary applications in mining and tunneling

4.1.1. Wudong coal mine

The Wudong coal mine is situated 34 km northeast of Urumqi, China. The coal seam has a thickness of 37.5 ± 5.6 m and is buried at a depth of approximately 450 m. It is entirely located in the south wing of the Badaowan syncline, with a dipping angle of 87°, making it a nearly vertical coal seam, as illustrated in Fig. 11(a). In the original support scheme, shown in Fig. 11(b), 20 mm diameter steel bolts were used as the primary support, while 18.9 mm diameter cable bolts were employed as the secondary support. [49].

Steel bolts caused operational delays and safety concerns due to the need for manual removal to prevent frictional sparks during the high-speed shearer cutting process. This manual removal was labor-intensive, risky, and negatively impacted mining productivity. To address these issues, high-strength 27 mm diameter FRP bolts were introduced as a safer and more efficient rib support solution. The FRP rockbolts offered several advantages, including being cuttable without generating sparks and being lightweight for easier handling.

Laboratory tests demonstrated that the FRP bolts had a tensile strength of 486 MPa and a shear strength of 258 MPa. Field tests revealed that the peak anchorage capacities of FRP bolts were 31 % higher than those of steel bolts. Fully grouted FRP bolts were used to mitigate shear deformation weaknesses and enhance anchorage

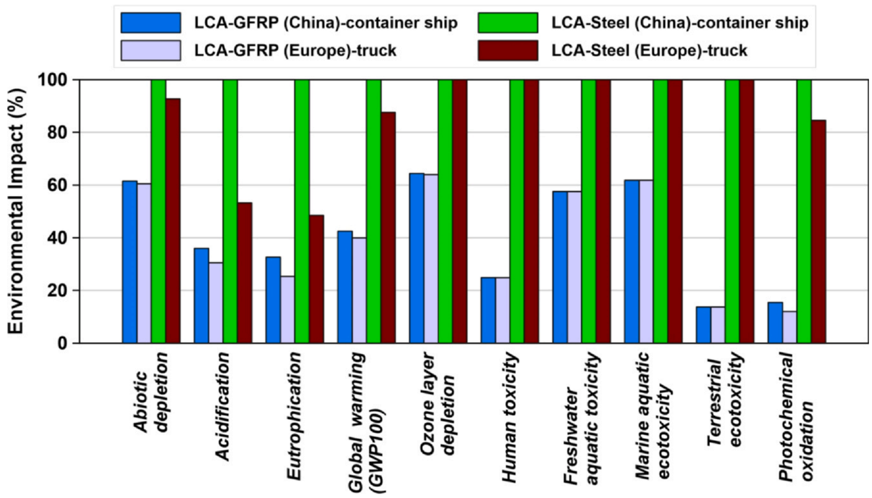

Fig. 10. Characterization of a rockbolt – shipped from China and Europe (adopted from Kodymova et al. [66]).

performance. FLAC3D simulations showed that 27 mm diameter FRP bolts reduced rib deformation more effectively than steel bolts, achieving a horizontal displacement reduction of up to 38 % [49].

The implementation of 27 mm diameter FRP bolts enhanced operational safety, minimized deformation in large deformation roadways, and boosted mining productivity at the Wudong coal mine. This application underscores the effectiveness of large-diameter FRP bolts for deep underground coal mining and other high-stress environments.

4.1.2. Application of FRP bolts in monitoring the internal force of rocks surrounding a Mine-shield tunnel

In a project involving a 2.5 km long railway tunnel in the northwest of the Pearl River Delta, China, FRP rockbolts were used to monitor the internal forces of the surrounding rocks. This tunnel was constructed in an acidic environment with high concentrations of ammonia and nitrogen, making it an ideal candidate for FRP rockbolts due to their high tensile strength, low shear strength, and excellent corrosion resistance [18].

The FRP rockbolt, as shown in Fig. 12**(a)**, was 8.2 m long with a diameter of 18 mm. It had a tensile capacity of 203 kN, a shear capacity of 101 kN, and a modulus of elasticity of 45 GPa. The rockbolt was equipped with an anchor head dynamometer to measure the internal forces, as shown in Fig. 12**(b)**. Additionally, solar energy collection technology was integrated to create a remote monitoring system. The study found that FRP rockbolts effectively responded to events such as blasting vibrations and construction disturbances, providing clear patterns for monitoring the internal forces in the surrounding rock [18]. This case study highlights the advantages of using FRP rockbolts in acidic environments, demonstrating their effectiveness in maintaining structural integrity and providing reliable monitoring data.

4.2. Permanent applications in mining and tunneling

4.2.1. Jurong Rock Caverns in Singapore

GFRP rockbolts, including solid and hollow (tubular) types, were chosen for ground control in the Jurong Rock Caverns (JRC) in Singapore, the first underground rock-cavern facility in Southeast Asia to store liquid hydrocarbons. This application served as an alternative to conventional steel rockbolts due to the corrosive hydrologic environment, considerations for weight and handling, local construction preferences, and the requirement for a 50-year cavern design life. The GFRP rockbolts were manufactured using vinyl-ester resin and ECR-glass fibers. Over 80,000 cement-grouted GFRP solid and tubular bars, ranging from 3.5 to 5.5 m in length, were implemented in the project.

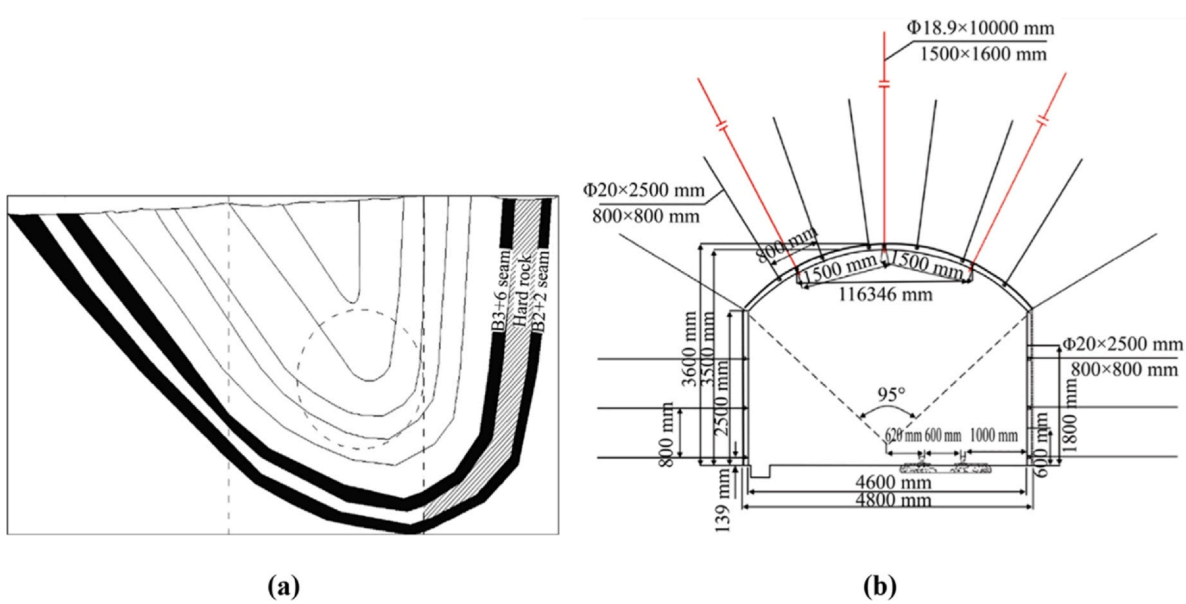

Fig. 11. (a) Geological profile; and (b) original support scheme in Wudong coal mine [49].

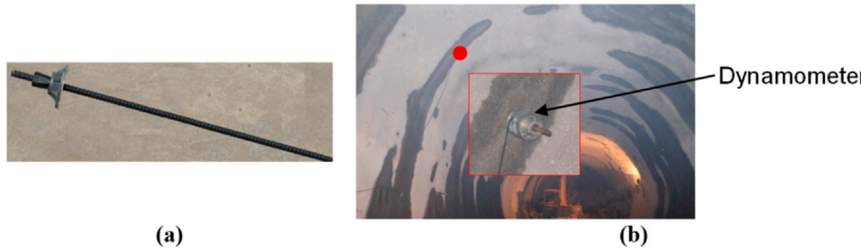

Fig. 12. (a) FRP rockbolt; and (b) Field deployed with dynamometer [18].

A comprehensive test program assessed the long-term durability of these GFRP rockbolts in a saline solution simulating the subsea cavern water. The study predicted a high tensile strength retention of approximately 97 % after 50 years and greater than 82 % even after 100 years at Singapore's mean annual temperature of 32 °C. These findings confirmed the suitability of GFRP rockbolts as a long-term solution for ground control in the Jurong Rock Caverns [15].

4.2.2. Sydney Metro Northwest tunneling project

The application of GFRP rockbolts was investigated in the Sydney Metro Northwest tunneling project, where pull tests were conducted on GFRP threaded bar bolts using both resin and cement grout in the sandstones and shales of the Sydney region. The pull test results revealed a wide range of stiffness values for GFRP bolts, showing both stiff and flexible behavior similar to steel bolts. Notably, GFRP bolts cement-grouted in shale exhibited stiff performance. The study concluded that there was no significant difference in pull-out stiffness or grout-bar bond strength between steel and GFRP rockbolts [57]. Overall, the investigation suggests that GFRP bolts can be a viable option for rock reinforcement, demonstrating comparable performance to traditional steel bolts in the geological conditions of the Sydney Metro Northwest project.

4.2.3. Vereina Tunnel, Switzerland

The Vereina Tunnel, a 19 km long railway tunnel in Switzerland, provides a compelling case study for the successful application of FRP rockbolts. The tunnel passes through diverse rock formations, including highly fractured and unstable zones. Significant water ingress was encountered during construction, posing a risk of corrosion to traditional steel bolts. Additionally, the tunnel was designed for a long service life, requiring a support system that could withstand long-term degradation. GFRP rockbolts were selected as a primary support element due to their high strength, corrosion resistance, and long-term durability [16]. GFRP rockbolts were used in conjunction with other support measures, such as shotcrete and steel arches, to provide comprehensive ground stabilization (Fig. 13).

The Vereina Tunnel was successfully constructed, with the GFRP rockbolts providing effective and reliable ground support. After more than 20 years of service, the GFRP rockbolts have shown excellent performance with no signs of significant degradation or corrosion. The corrosion resistance of GFRP bolts has resulted in minimal maintenance requirements, reducing operational costs [16].

This case study demonstrates the suitability of GFRP rockbolts for challenging tunneling projects with complex geological conditions and water ingress. The long-term performance of GFRP rockbolts in the Vereina Tunnel highlights their durability and cost-effectiveness compared to traditional steel bolts in corrosive environments. The successful application of GFRP rockbolts in this major infrastructure project has contributed to the wider acceptance and adoption of FRP technology in tunneling and mining. This case study exemplifies how FRP rockbolts can provide a reliable and long-lasting solution for ground support in demanding environments, offering significant advantages over traditional steel alternatives.

FRP rockbolts have been utilized in both temporary and permanent applications in mining and tunneling. Their development for corrosion-free long-term anchoring opens up new application fields for the future [17,52], such as permanent support for corrosive underground structures, areas with heavy mine water flows causing acid mine drainage, areas with dynamic loading, and regions experiencing frost-thaw cycles. The next section will discuss the knowledge gaps and research requirements for the long-term performance of FRP rockbolts.

Fig. 13. Ground support in squeezing conditions using GFRP rockbolts and shotcrete [16].

5. Knowledge gaps and future directions

Research on FRP rockbolts has significantly expanded our understanding of their behavior and potential in ground support applications. Studies have extensively investigated the mechanical properties of various FRP composites, including tensile and shear strength, under different loading conditions [52,69–71]. Recent tests have focused on the use of FRP rockbolts for geotechnical engineering applications [52] and tunnel face reinforcement [72], supporting their use in such applications. However, research on FRP rockbolts in low pH conditions is limited.

Long-term durability studies aim to conduct extensive field research to monitor the performance of FRP rockbolts in low pH environments over extended periods. The objective should be to understand the degradation mechanisms and lifespan of these rockbolts under acidic conditions. Studies can use accelerated testing methods, such as immersing FRP samples in acidic solutions at elevated temperatures [73]. These tests can provide insights into the long-term performance of FRP rockbolts in a relatively short time. However, field studies are essential to validate the long-term performance of FRP rockbolts in real-world conditions. This knowledge will be crucial in designing more reliable and durable ground support systems, ensuring enhanced safety and cost-effectiveness in challenging mining and tunneling applications.

In the literature, most environmental impact assessments focus on FRP rebars in the construction industry [74,75]. To date, only a single published LCA study exists on the use of FRP bolts in a tunnel project in Scandinavia [66]. An environmental impact assessment should aim to evaluate the benefits of using FRP rockbolts compared to traditional steel bolts, with a focus on lifecycle analysis and sustainability. This will help in understanding how FRP rockbolts contribute to reduced environmental footprints through factors such as lower energy consumption during production, reduced transportation emissions due to their lightweight nature, and enhanced durability leading to fewer replacements. Highlighting these environmental advantages can promote the wider adoption of FRP rockbolts in the mining industry, encouraging more sustainable practices and reducing the overall ecological impact of mining operations.

Liu et al. [18] evaluated the effectiveness of FRP rockbolts in monitoring the internal forces of surrounding rocks during tunnel construction, demonstrating a safe and effective method for remote monitoring of the mechanical behavior of rocks around a mine-shield tunnel. However, the research was constrained to a single monitoring point due to site condition limitations. Therefore, future research should focus on integrating advanced sensing technologies into FRP rockbolts to enable real-time monitoring of internal forces in different environmental conditions. This involves embedding sensors within the rockbolts to continuously collect data on stress, strain, temperature, and other critical parameters. The anticipated impact of this research is significant, as it would provide valuable data for proactive maintenance, allowing for timely interventions before issues escalate. Additionally, it would enhance the overall safety of mining operations by offering insights into the structural integrity of the support system and environmental conditions, thereby preventing potential failures and ensuring a safer working environment.

6. Conclusions

This study presents a detailed review of fiber-reinforced polymer (FRP) rockbolt technology and corrosion issues in low pH mines. A comparison is made between the material properties of conventional and FRP rockbolts. The long-term application of FRP rockbolts is assessed. The following conclusions can be drawn from the study. First, FRP rockbolts offer several advantages over traditional steel rockbolts, including high tensile strength, greater flexibility, lighter weight, excellent fatigue resistance, and superior corrosion resistance. These properties make them highly desirable for reinforcing rock excavations, particularly in harsh environments with low pH mines (pH $<$ 5). The reviewed case studies demonstrate the success of FRP rockbolt in both temporary and permanent applications. However, to further enhance the knowledge of FRP rockbolts and their potential use in low pH mines, long-term durability studies are encouraged. Also, the environmental impact assessment should aim to evaluate the benefits of using FRP rockbolts compared to traditional steel rockbolts, with a focus on lifecycle analysis and sustainability. Future research should focus on integrating advanced sensing technologies into FRP rockbolts to enable real-time monitoring of internal forces in harsh environmental conditions.

CRediT authorship contribution statement

Wenxue Chen: Writing – review & editing, Writing – original draft, Methodology. **Girish Narayan Prajapati:** Writing – review & editing, Writing – original draft. **Hani S. Mitri:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Wenxue Chen is currently employed by SFTec Inc. All other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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